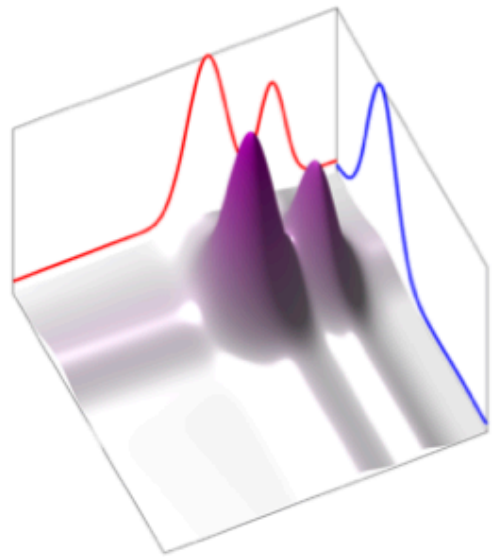
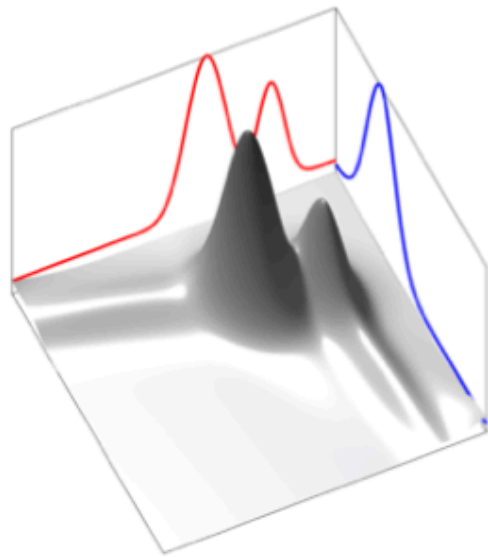
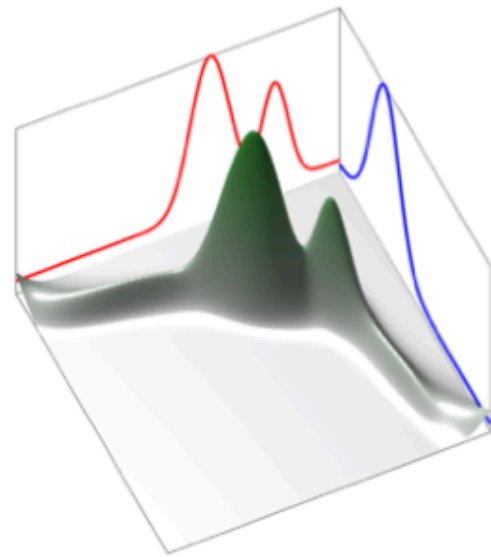




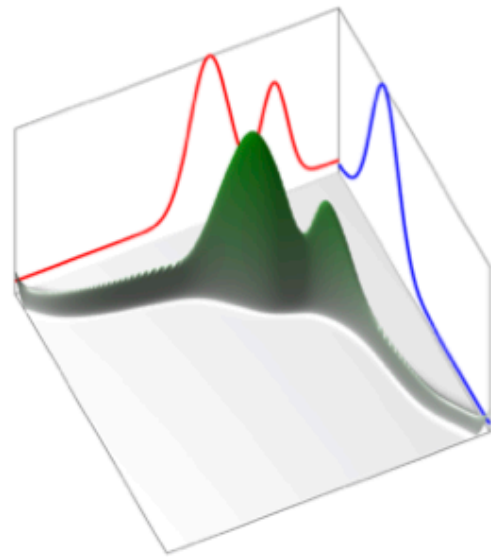
$$\varepsilon = 10$$



$$\varepsilon = 1$$



$$\varepsilon = 10^{-1}$$

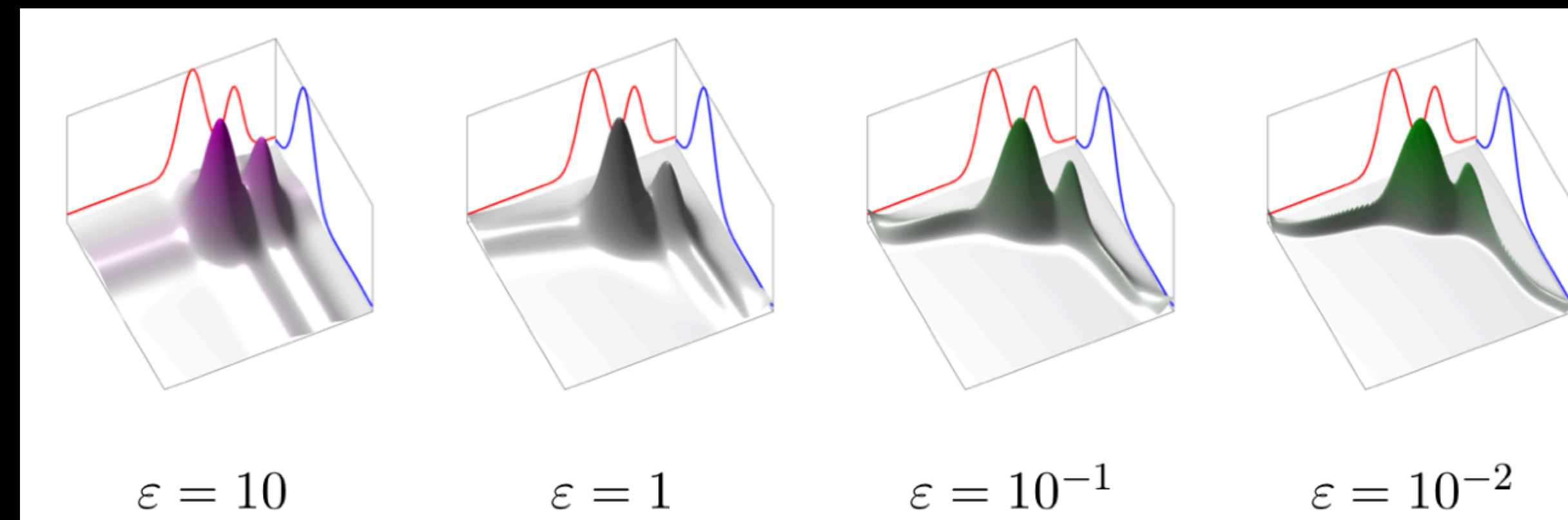


$$\varepsilon = 10^{-2}$$

Entropic OT Formulation

Regularized problem: $\min_{\pi \geq 0} \langle C, \pi \rangle - \varepsilon H(\pi), \quad H(\pi) = - \sum_{i,j} \pi_{ij} \log \pi_{ij}$

- $\varepsilon > 0$: regularization strength.
- Trade-off:
 - Small $\varepsilon \rightarrow$ close to true OT, but less smooth.
 - Large $\varepsilon \rightarrow$ more blur, faster computation.
- Properties:
 - Strongly convex $\rightarrow$ unique solution.
 - Differentiable w.r.t. inputs (α, β, C) .



Entropic OT Formulation